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Example Conversation

Customer: "I am moving next month and want to cancel unless fiber is available at my new place. Can you transfer my service once availability is confirmed and also waive any install fee?"
VA: "I can certainly help with that. To ensure I have the most accurate details, could you please provide your account number?"
VA: "Thanks, Muru. I see you've been with us for 4.2 years—we really appreciate you being with us!"
VA: "I'll check fiber availability at your new location and guide you through the move. What is the new address you're moving to, and what is your target move date?"
Customer: "100 First Ave, Dallas, TX 75202. Moving next Friday."
VA: "Got it. I'll check service at 100 First Ave, Dallas, TX 75202 for Friday, May 12."
VA: "Before I run the check—if fiber is available, are you open to upgrading from Internet 500 to Fiber 1 Gbps for $80/month? I can also check if you are eligible for an installation fee waiver."
Customer: "Yes, let's do 1 Gbps. Please waive the install fee."
VA: "Checking now... Great news—fiber is available at that address!"
VA: "We can upgrade you to Fiber 1 Gbps. Since your account is month-to-month, there is no early termination fee even if you were to cancel."
VA: "Let's schedule your installation. I have appointments available on May 13 AM, May 13 PM, and May 14 AM. Which works for you?"
Customer: "May 13 AM works."
VA: "Perfect—your Fiber 1 Gbps install is scheduled for May 13 between 8–12 AM."
VA: "Regarding the $99 install fee: You qualify for a mover's fee waiver because you have 3+ years of tenure, no waivers in the last 12 months, and Autopay is active. I can apply that now."

Customer: "Please apply it. Thank you."
VA: "Applied—the fee has been waived and noted on your order."
VA: "Before we finalize, I see an outstanding balance of $82.45 on your current account at 100 First St. We need to clear this balance to proceed with the connection at the new address. Would you like to pay that now using the card on file?"
Customer: "Yes, please pay it using the card on file."
VA: "Got it. Processing the payment of $82.45 from your Visa ending in 3026... ✅ Payment successful. Your current balance is now $0.00, and I'll include the receipt in your email confirmation to muru.a@mail.com."
VA: "Now that the balance is cleared, I'll finalize the move order. This requires disconnecting the service at your current address. Shall I schedule that disconnect for your move date, May 12?"
Customer: "Yes, disconnect the old service on May 12. Start the new Fiber 1 Gbps on May 13."
VA: "Perfect. I'll disconnect service at 100 First St on May 12 and activate fiber at your new address on May 13 AM. Your payment is cleared and I have sent a prepaid return label to your email. **Please use this to return your old gateway via mail, as the technician cannot take the equipment back.**"
Customer: "Sounds good."
VA: "Here's a quick summary: • **Outstanding bill:** $82.45 paid via Visa ending 3026 • **Disconnect:** May 12, Old Address (100 First St) • **New install:** May 13, 8–12 AM, Fiber 1 Gbps, install fee waived • **Monthly:** $80 • **Confirmation:** Sent to muru.a@mail.com and via SMS • **Return label:** Sent Would you like me to send a reminder the day before to prep for install and check ONT access?"
Customer: "Yes, please remind me."
VA: "Reminder set. I'll notify you on May 12 at 10 AM with a prep checklist. Anything else I can help you with today?"

Requirements Document

1. Purpose and Scope

1.1 Product & Account Scope

- **Product:** Strictly **Home Internet only** (Copper Internet or Fiber Internet).

- **Customer Constraints:**
  - **Existing Customers:** One account per customer; one active service location per account.
  - **New Customers:** Supported via an "Unauthenticated" flow (No Account ID required). The system must allow users to query availability and "sign up" for new service.

1.2 Geographic Scope

- **Region:** Services are only available in the fictional **"Metro City"** region (Zip Code 75202).
- **Address Constraints:** The world relies on a **finite, hardcoded list of valid addresses (~20 total)** divided into specific use cases:
  - **Existing Customer Addresses (~10):** A distinct set of addresses where current personas live.
  - **Destination/New Addresses (~10):** A distinct set of vacant addresses used specifically for "Move" requests or "New Connection" sales.
- **Validation:** Any address outside these specific lists is treated as "out of footprint", and service must be denied.

1.3 Functional Scope (In-Scope/Out-of-Scope)

- **In-Scope (The "Happy Paths"):**
  - **Moves:** Transferring existing service to a new address.
  - **New Connections:** Sales flow for users without an Account ID. Users must provide a valid "Destination/New" address existing in the database to proceed.
  - **Plan Changes:** Upgrades/Downgrades based on address capability.
  - **Fiber Eligibility:** Checking max speed availability at a specific location.
  - **Financials:** Balance inquiries, one-time payments, and logic-driven fee waivers.
  - **Scheduling:** Appointment booking **strictly** for new installations (New Customers or Moves) where the target address lacks equipment.
  - **Account Cancellation:** Cancellation of service at the current address. **Constraint:** The Agent must ensure the account balance is paid off ($0.00) before processing the cancellation.
- **Out-of-Scope:**
  - Wireless/Mobile, TV, bundles, and promotions.

    - Repair/Troubleshooting appointments (service is assumed to be working).
    - Collections/Legal flows.

1.4 Temporal & Interaction Logic

- **Time:** The Agent must understand relative dates (e.g., "Next Friday") but assumes **infinite appointment capacity** (all future AM/PM slots are available).
- **Appointment Logic:**
    - **IF** Equipment is present $\to$ **Self-Install** (No Appointment).
    - **IF** Equipment is missing $\to$ **Technician Appointment Required**.
- **Interaction Rule:** All out-of-scope requests must be declined gracefully with a redirection to a human agent or a generic *"I cannot help with that"* message.

2. Global World Assumptions

2.1 Accounts and Products (Global constraints)

- **Account Constraints:**
    - **One-to-One Relationship:** Each customer is limited to **at most one account**.
    - **Service Type:** Accounts are strictly **Internet-only** (No TV, Voice, or Wireless bundles).
- **Product Catalog, Speeds & Pricing:**
    - The Agent must use the following fixed pricing schedule for all quotes and billing calculations. All prices are flat rates (taxes excluded).

Technology	Plan Name	Speed (Down/Up)	Monthly Price
Copper	Internet 100	100 Mbps	**$45.00**
Fiber	Fiber 300	300 Mbps	**$55.00**
Fiber	Fiber 500	500 Mbps	**$65.00**
Fiber	Fiber 1 Gig	1000 Mbps	**$80.00**
Fiber	Fiber 2 Gig	2000 Mbps	**$110.00**

- **Plan Terms (Financial Rules):**
    - **Contract Status:** All plans are strictly **Month-to-Month (MTM)**.
    - **Commitment:** There are **no contracts**, **no term commitments**, and **no Early Termination Fees (ETFs)**.
- **Account Status Logic:**

    - The database supports only two account statuses:
        - **ACTIVE:** The customer has live service and can perform moves, upgrades, or payments.
        - **CANCELED:** The customer has disconnected service.
    - **Restoration Rule:** Once an account status is set to **CANCELED**, it **cannot be restored** or reactivated.
        - *Agent Action:* If a user authenticates with a CANCELED account, the Agent must explain the status and guide the user to either provide a different Active ID or start a "New Connection" sales flow.
- **Technology Migration Rules:**
    - **Copper Legacy:** Customers currently on Copper can continue with Copper at a new address (all 20 addresses are eligible for Copper) or upgrade to Fiber (if available).
    - **Fiber Upgrade:** The Agent must prioritize offering Fiber upgrades whenever a Copper customer moves to or queries a Fiber-eligible address.

2.2 Address Universe and Serviceability (The "Master Address List")

2.2.1 The Master Address List (20 Fixed Records)

- The Agent recognizes **only** the following 20 addresses. Any input outside this list returns an "Out of Footprint" error.
- **Group A: Current Customer Addresses (Occupied)**
- `100 First Ave, Dallas, TX 75202`
- `200 Second Ave, Dallas, TX 75202`
- `300 Third Ave, Dallas, TX 75202`
- `400 Fourth Ave, Dallas, TX 75202`
- `500 Fifth Ave, Dallas, TX 75202`
- `600 Sixth Ave, Dallas, TX 75202`
- `700 Seventh Ave, Dallas, TX 75202`
- `800 Eighth Ave, Dallas, TX 75202`
- `900 Ninth Ave, Dallas, TX 75202`
- `1000 Tenth Ave, Dallas, TX 75202`
- **Group B: New / Destination Addresses (Vacant)**
- `100 First St, Dallas, TX 75202`
- `200 Second St, Dallas, TX 75202`
- `300 Third St, Dallas, TX 75202`

- `400 Fourth St, Dallas, TX 75202`
- `500 Fifth St, Dallas, TX 75202`
- `600 Sixth St, Dallas, TX 75202`
- `700 Seventh St, Dallas, TX 75202`
- `800 Eighth St, Dallas, TX 75202`
- `900 Ninth St, Dallas, TX 75202`
- `1000 Tenth St, Dallas, TX 75202`

2.2.2 Serviceability Data Attributes

- For every address above, the database must store the following serviceability flags to drive Agent logic:
  - `max_tech_type`: The highest technology available (Values: `Fiber` or `Copper`).
  - `max_speed`: The specific top speed tier (Values: `100`, `300`, `500`, `1000`, `2000`).
  - `equipment_installed`: Boolean flag indicating if an ONT/Jack is present (TRUE = Self-Install, FALSE = Technician Required).

2.2.3 Database Population Rules (Scenario Generation)

- To ensure all demo scenarios (Upgrades, Constraints, Appointments) are testable, the database generation script must distribute attributes across the 20 addresses as follows:
  - **The "Fiber Ready" Cohort (Approx. 50% of addresses):**
    - **Settings:** `max_tech_type = Fiber`, `max_speed = 1000` (1 Gig) or `2000` (2 Gig).
    - **Purpose:** Enables the "Fiber Upgrade" upsell flow.
  - **The "Copper Constraint" Cohort (Approx. 30% of addresses):**
    - **Settings:** `max_tech_type = Copper`, `max_speed = 100`.
    - **Purpose:** Forces the Agent to explain technology limitations ("Sorry, only 100 Mbps is available here").
  - **The "Technician Required" Cohort:**
    - **Settings:** Ensure at least 5 addresses (mixed Fiber/Copper) have `equipment_installed = FALSE`.
    - **Purpose:** Forces the Agent to trigger the "Appointment Scheduling" flow.
  - **The "Self-Install" Cohort:**

        - **Settings:** Ensure remaining addresses have `equipment_installed = TRUE`.
        - **Purpose:** Allows the Agent to skip scheduling and offer immediate activation.

2.3 Time and Move Dates

2.3.1 Move Date / Start Date Validation

- The Agent must validate the user's requested date against the following hard constraints:
    - **Validity:** The input must be resolvable to a valid calendar date.
    - **No Past Dates:** The date cannot be in the past.
    - **Minimum Lead Time:** The date **cannot** be the same day as the current date (Today).

2.3.2 Move & Install Scheduling Logic

- **Availability Model:**
    - Available slots are **not** stored in a central database (Infinite Capacity model).
    - All slots are considered "empty" and available to begin with.
    - **Locking:** Once a slot is selected by the customer, it is recorded in the conversation state/database as "locked" to that specific order.
- **Slot Definitions:**
    - **AM Slot:** 8:00 AM – 12:00 PM.
    - **PM Slot:** 1:00 PM – 5:00 PM.
- **Presentation & Selection Rules:**
    - **Initial Offer:** The VA should offer exactly **4 options** initially (e.g., "I have tomorrow AM, tomorrow PM, or the day after AM available").
    - **Expansion:** The VA provides 4 more options **only** if the user explicitly asks for them.
    - **Strict Selection:** Customers can **only** pick from the AM/PM slots offered. Specific times (e.g., "9:30 exactly") are **not supported**.

3. Authentication and Identity Rules

- Objective: Define a strict, deterministic method for user identification using a simplified 5-digit Account ID.

3.1 Authentication Method (Account ID Only)

- **Primary Key:** The Agent relies **exclusively** on a 5-digit `Account_ID`.
- **Format:** Continuous digits (e.g., `10001`). No dashes or spaces allowed.
- **Unsupported Identifiers:** The Agent must **not** attempt to look up accounts by Name, Phone Number, Email, Address, or Password.
    - *Agent Response to Unsupported Input:* "For security, I can only access your account with your 5-digit Account ID. Please provide that number.".
- **New Customers:** Users without an Account ID are treated as "New Customers" and routed to the Sales / Availability Check flow.

3.2 Valid / Invalid Account IDs

- The Agent must validate the provided ID against the database on every attempt:
    - **Scenario A: ID is Valid & ACTIVE**
        - **Action:** Authentication **Success**.
        - **Result:** The Agent loads the customer context (Name, Address, Plan) and proceeds with the requested journey.
    - **Scenario B: ID is Valid but CANCELED**
        - **Action:** Authentication **Partial**.
        - **Result:** The Agent explains the status: *"I see that account [ID] is currently canceled/closed."*
        - **Next Step:** The Agent asks if the user has a different active ID or wants to set up a new connection.
    - **Scenario C: ID is Invalid (Not in Database)**
        - **Action:** Authentication **Failure**.
        - **Result:** The Agent prompts for a retry: *"I could not find an account with ID [Input]. Please check the number and try again."*.
    - **Retry Limits (Error Handling)**
        - **Rule:** After **2 consecutive failed attempts** (invalid IDs).

        - **Action:** The Agent stops asking for the ID to avoid a loop.
        - **Response:** *"I'm having trouble locating that account. I can still answer general questions about our speeds and pricing. How can I help?"*.

3.3 Context-Aware Greeting Logic

- Immediately upon successful authentication (Scenario A), the Agent must personalize the greeting based on the customer's tenure.
- **High Tenure (≥ 3 Years):**
    - *Logic:* Acknowledge loyalty.
    - *Script:* "Welcome back, [Name]. I see you've been with us for [Tenure] years—thank you for your loyalty! How can I help?"
- **Standard/New Tenure (< 3 Years):**
    - *Logic:* Standard polite welcome.
    - *Script:* "Hi [Name], thanks for verifying your account. How can I help you today?"

4. Customer Personas & Scenario Data

- **Objective:** Define the user population. Instead of random data, we use specific "Archetypes" designed to trigger specific demo scenarios.

4.1 The "Cheat Sheet" Strategy (20 Accounts)

- **Fixed Data Set:** The database will contain exactly **20 distinct accounts**.
- **ID Format:** 5-digit integers ranging from **10001** to **10020**.
- **Demo Logic:** The End User selects an ID from the "Cheat Sheet" to test a specific logic path (e.g., "I want to test the Fee Waiver logic, so I will use ID 10001").

4.2 Persona Archetypes (Scenario Mapping)

- To ensure all business rules are testable, the 20 accounts must be distributed across the following logic buckets:

- **Archetype A: The "Perfect" Candidate (Waiver Success)**
  - **Profile:** Active, Tenure > 3 Years, AutoPay = TRUE, Balance = $0.00.
  - **Demo Outcome:**
    - **Fee Waiver: PASS** (Agent grants $0 install fee).
    - **Move:** Proceeds immediately (No billing gate).
- **Archetype B: The "New" Customer (Waiver Failure)**
  - **Profile:** Active, Tenure < 3 Years (e.g., 0.5 years), AutoPay = FALSE, Balance = $0.00.
  - **Demo Outcome:**
    - **Fee Waiver: FAIL** (Agent enforces $99 install fee).
- **Archetype C: The "Debtor" (Billing Gate)**
  - **Profile:** Active, Balance > $0.00 (e.g., $82.45).
  - **Demo Outcome:**
    - **Move Request: BLOCKED**. Agent pauses flow and demands payment before proceeding.
- **Archetype D: The "Legacy" User (Upsell Target)**
  - **Profile:** Active, Current Plan = Copper 100.
  - **Demo Outcome:**
    - **Fiber Eligibility:** Agent identifies the user is on old tech and aggressively pitches the Fiber upgrade if the new address supports it.
- **Archetype E: The "Churned" User (Status Check)**
  - **Profile:** Status = **CANCELED**.
  - **Demo Outcome:**
    - **Login:** Agent recognizes the user but refuses "Active" workflows, offering a New Connection flow instead.

4.3 Data Reset Requirements

- **Requirement:** The system must support a "Reset State" command (e.g., a SQL script `reset_world.sql`).
- **Function:** This command restores all 20 accounts to their original addresses, balances, and plan types.
- **Why:** Ensures that if a demo user moves Account 10001 to a new house, the next demo session starts with Account 10001 back at their original home.

5. Journeys in Scope

5.1 Existing Customer – Authenticated Journeys

5.1.1 Move Internet Service to New Address

- **Precondition:** Customer provides a **valid, ACTIVE** Account ID.
- **Balance Gate:** If `pending_balance > 0`, the customer **must pay it off** before the move can proceed.
- **Logic Flow:**
  - **Capture New Address:** Must be in the supported "Destination" list.
  - **Validate Date:** Future date required; same-day moves not allowed.
  - **Check Eligibility:**
    - *If Fiber Available:* Proactively offer Fiber upgrade.
    - *If Fiber Not Available:* Offer "Lateral Move" (Keep Copper 100) or Cancel.
  - **Finalize:** Close old account (Status=`CANCELED`) and create new account (Status=`ACTIVE`) on the move date.

5.1.2 Cancel Internet Service

- **Precondition:** Valid, ACTIVE Account ID.
- **Action:**
  - Ensure `pending_balance = $0.00`.
  - Set account status to **CANCELED**.
  - *Constraint:* Once canceled, accounts cannot be restored.

5.1.3 Plan Changes (Current or New Address)

- **Scenario A (Upsell):** Customer wants 1 Gig, and address supports it $\to$ **Grant Upgrade**.
- **Scenario B (Downgrade/Constraint):** Customer wants 1 Gig, but address only supports 500M $\to$ **Offer Max Available Tier**.
- **Scenario C (Price Downgrade):** Customer wants to lower bill $\to$ **Support Downgrade** if lower tier exists.

5.1.4 Reschedule Appointment

- **Action:** Modify an existing technician appointment to a new future date/time slot.
- **Constraints:** New date must follow standard validation rules (Future date, AM/PM slots only).

5.2 New Customer – Unauthenticated Journeys (Sales)

- **Precondition:** No Account ID required.
- **Logic Flow:**
  - **Capture Address:** Must be a valid "New/Destination" address from DB.
  - **Check Serviceability:** Determine if Fiber or Copper is available.
  - **Offer Plans:**
    - *If User mentions "Fiber":* Show eligible Fiber tiers.
    - *If User is vague:* Show "Best Available" plan.
  - **Create Account:** Generate new DB record with `Status = ACTIVE`, `Pending_Balance = 0`, and captured details.

5.3 Supported Micro-Actions (Standalone Capabilities)

- The VA must support these single-turn requests without forcing a full flow:
  - **Check Speed:** "What is the max speed here?" (Query `max_speed` column).
  - **Check Waiver:** "Do I qualify for a free install?" (Run 3-rule check).
  - **Pay Bill:** "Pay my balance." (Clear `pending_balance`).
  - **Update Contact Info:** "Actually, use my new email." (Update email or phone number in profile).

6. Input Validation Rules

6.1 Address Input & Matching Logic

- The Agent must strictly validate all geographic inputs against the "Master Address List" defined in Section 2.2.
  - **Strict Matching:**
  - **Rule:** The user's input must resolve to one of the ~20 hardcoded addresses in the database.

    - **Inference Rule (Partial Match):** Since the entire supported world is located in "Metro City," the Agent **must** accept partial street addresses (e.g., "100 First Ave") and automatically infer the City, State, and Zip (Dallas, TX 75202) for the lookup.
    - **Failure:** If the address is not found (e.g., "123 Mars Ave"), the Agent must return the "Out of Footprint" error, explain that serviceability cannot be checked, and offer to cancel the current account instead.
- **Occupancy Validation (The "Occupied" Trap):**
    - **Rule:** The destination address must be flagged as `Vacant` (Group B) in the database.
    - **Failure:** If the user provides an address that is already `Occupied` (Group A), the Agent must decline: *"I recognize that address, but there is already an active service account at that location. I can only set up service at a vacant address. Do you have a different address?"*
- **The "Same Address" Logic Trap:**
    - **Rule:** The Agent must compare the provided `New_Address` against the authenticated customer's `Current_Address`.
    - **Validation:** `New_Address` $\neq$ `Current_Address`.
    - **Error Handling:** If they match, the Agent must correct the user: *"It looks like you entered your current address as the new destination. Please provide the different address where you are moving to."*

6.2 Move/Start Date Validation

- The Agent parses date inputs to ensure logistical feasibility.
- **Date Quality Constraints:**
    - **Format:** Must be a recognizable calendar date (e.g., "May 12", "Next Friday").
    - **Future Only:** Dates in the past (e.g., "Yesterday") are **Invalid**.
    - **Lead Time:** The date **cannot** be "Today" (Current Date).
- **Slot Selection Constraints:**
    - **Valid Inputs:** Users can only select **AM** (Morning) or **PM** (Afternoon).
    - **Invalid Inputs:** Specific times (e.g., "9:30 AM" or "10:00 sharp") are **not supported**.
    - **Agent Correction:** If a user requests a specific time, the Agent must guide them back to the window: *"I can't schedule specific times, but I can book the 8-12 AM arrival window. Does that work?"*.

7. Serviceability & Sales Logic (The "Serviceability Agent")

7.1 Fiber Availability & Speed Provisioning

- The Agent must query the max_tech_type and max_speed columns for the target address to determine what can be sold.
- **The "Ceiling" Rule:** The Agent can **never** offer a plan faster than the address's `max_speed`.
    - *Example:* If Address A has `max_speed = 500`, the Agent **cannot** offer Fiber 1 Gig.
- **Proactive Checking:**
    - **Moves:** The Agent automatically checks availability as soon as the address is captured to see if an upgrade is possible.
    - **Inquiries:** If a user explicitly asks "Is Fiber available?", the Agent runs the check immediately.

7.2 Dynamic Plan Negotiation (Upsell/Downsell)

- The Agent adapts its offer based on the customer's reaction and the address's capabilities.
- **Scenario A: The Upsell (Copper $\to$ Fiber)**
    - *Trigger:* Customer is currently on Copper (or New) AND Target Address is Fiber-Ready.
    - *Action:* The Agent **must** pitch the upgrade: *"Great news, your new home is Fiber-ready. I can upgrade you to Fiber 1 Gig."*.
- **Scenario B: The Downsell (Price Sensitivity)**
    - *Trigger:* Customer rejects the top-tier offer (e.g., "That's too expensive").
    - *Action:* The Agent retrieves the full list of available lower tiers (e.g., Fiber 300, Fiber 500) and offers a cheaper option.
- **Scenario C: The Lateral Move (1:1 Transfer)**
    - *Trigger:* Target Address supports the customer's *current* speed exactly.
    - *Action:* The Agent offers to simply transfer the existing plan without changes.

7.3 Infrastructure Compatibility (Fiber vs. Copper)

- **The Copper Constraint:**
    - *Condition:* Target Address is `max_tech_type = Copper` (Max 100 Mbps).

- *Action:* If the customer requests Fiber, the Agent must explicitly state: *"Fiber is not available at this specific address. The fastest speed I can offer here is Internet 100."*.
- *Outcome:* If the user refuses the Copper speed, the Agent proceeds to the **Cancel** flow.

8. Scheduling & Appointment Logic (The "Scheduling Agent")

8.1 Appointment Triggers (Equipment Check)

- The Agent determines if a technician is required based on the destination address's hardware status.
- **Self-Install (No Appointment):**
  - **Condition:** Destination address has `equipment_installed = TRUE`.
  - **Action:** Agent offers immediate activation on the requested date. No time slot selection is needed.
- **Technician Install (Mandatory Appointment):**
  - **Condition:** Destination address has `equipment_installed = FALSE`.
  - **Action:** Agent **must** schedule a technician visit.

8.2 Slot Availability Rules (Infinite Capacity)

- **Capacity Model:** The system assumes "Infinite Capacity." It does not check a database for busy/free slots; all valid future slots are considered available.
- **Slot Definitions:**
  - **AM Slot:** 8:00 AM – 12:00 PM.
  - **PM Slot:** 1:00 PM – 5:00 PM.
- **Presentation Logic:**
  - **The "Rule of 4":** The Agent should offer exactly **4 options** initially (e.g., "I have tomorrow AM, tomorrow PM, or the day after AM").
  - **Expansion:** Additional 4 options are provided **only** if the user explicitly asks.
  - **Restrictions:** No past dates; no "Same Day" appointments.

9. Financial & Billing Logic (The "Billing Agent")

9.1 Balance Gating & Payments

- **The Move Gate:** A "Move" order **cannot** proceed if the account has a `pending_balance > $0.00`.
  - *Agent Action:* Pause the move flow, explain the balance, and require payment.
- **Payment Execution:**
  - **Method:** Payments are processed exclusively using the **Card on File**.
  - **Result:** Upon user confirmation, the Agent clears the balance to `$0.00` immediately to allow the journey to proceed.
- **Billing Constraints:**
  - **No Proration:** The new plan's **full price** applies immediately. Prorated charges are not supported.
  - **No Credits:** Negative account balances (Account Credits) are not supported.
- **Future Bill Inquiries:**
  - ***Logic:*** If asked about the next bill, the Agent states it will be posted on the **1st of the next month**.
  - ***Amount:*** The amount quoted is based strictly on the price of the **currently active plan**.

9.2 Fee Waiver Evaluation (The "3-Rule" Logic)

- The standard Installation Fee is **$99.00**. The Agent evaluates eligibility for a waiver ($0.00) based on strict criteria.
- **The Logic Gate:** The waiver is granted **IF AND ONLY IF** all three conditions are met:
  - **Tenure:** Account age must be **> 3 Years**.
  - **History:** No waivers applied in the last 12 months (derived from chat history/business rules).
  - **Payment: AutoPay** must be currently **Active**.
- **Outcome:**
  - *Pass:* Agent confirms the waiver: *"Good news, I can waive the installation fee."*
  - *Fail:* Agent enforces the charge: *"I checked your eligibility, but the $99 fee applies because..."*

9.3 AutoPay Handling

- **Status Check:** The Agent uses the `autopay_active` flag primarily as a criterion for the Fee Waiver logic.
- **Management:** The Agent is authorized to **Enable or Disable** AutoPay upon specific user request (updating the `autopay_active` status). However, complex banking updates (changing the Card on File) remain out of scope.

10. Order Configuration & Lifecycle

10.1 Move Order Logic (Disconnect Old / Connect New)

- Technically, a "Move" is executed as two distinct, linked transactions.
- **Workflow:**
  - **Create New Account:** The Agent creates a new record for the destination address with `Status = ACTIVE` starting on the **Activation Date**.
  - **Cancel Old Account:** The Agent sets the current account to `Status = CANCELED` effective on the **Disconnect Date**.
- **Date Synchronization:**
  - **Standard Rule:** To avoid service gaps, the **Disconnect Date** is set to **1 Day Prior** to the **Activation Date** (e.g., Disconnect May 12 $\to$ Activate May 13).
- **Billing Gate:** The Agent **cannot** finalize this order if the old account has a `pending_balance > 0`. The balance must be cleared first.

10.2 Cancellation Logic

- **Trigger:** Customer requests to stop service without moving.
- **Prerequisites:**
  - **Zero Balance:** Agent checks `pending_balance`. If >0, payment is mandatory before cancellation.
- **Irreversibility:** Once `Status` is set to `CANCELED`, it cannot be reversed. If the customer changes their mind later, they must start a "New Connection" flow.

11. Equipment Handling

11.1 Installation Requirements

- The Agent determines hardware needs based on the equipment_installed flag of the Destination Address (see Section 2.2).
- **If Equipment Installed = TRUE:**
  - **Type:** Self-Install.
  - **Action:** Agent instructs user to "Plug and Play" on the activation date. No Technician fee is added.
- **If Equipment Installed = FALSE:**
  - **Type:** Technician Install.
  - **Action:** Agent adds the **$99.00 Installation Fee** (unless waived) and mandates an appointment slot selection.

11.2 Incompatible Equipment (Technology Switches)

- **Scenario:** Customer moves between different infrastructure types (e.g., **Copper** $\to$ **Fiber**).
- **Logic:** The Agent must recognize that Copper modems are incompatible with Fiber ONTs.
- **Scripting:** The Agent must explicitly state: *"Your current equipment will not work with the new Fiber service. A technician will bring the new device."*

11.3 Return Logistics

- **Trigger:** Any Cancellation or Technology Switch move.
- **Constraint:** Old equipment must **ALWAYS** be returned via mail. Technicians **cannot** take back the old equipment during the install visit.
- **Action:** The Agent automatically triggers a "Return Label" notification.
- **Agent Script:** *"I have emailed a prepaid return label to [Email]. Please return your old equipment via mail to avoid unreturned equipment charges. Please note the technician cannot take the old equipment."*

12. Notifications, Summaries, and Reminders

12.1 Order Confirmation and Summary

- At the end of every successful transaction (Move, Payment, Sale), the Agent must generate a structured summary block to confirm details before closing.
- **Content:** The summary must include the following fields:
  - **Order Reference:** A generated, random 7-character alphanumeric string (e.g., `#ORD-8821X`) for professional realism. *Note: This ID is for display only and is not stored in the database.*
  - **Action Type:** (e.g., "Move to 100 First St" or "Payment Received").
  - **Effective Date:** The start date for new service or the disconnect date for cancellations.
  - **Financials:** Monthly recurring total and any one-time charges (or "Waived").
  - **Next Steps:** Instructions for equipment return or technician arrival.
- **Delivery:** The Agent confirms that a receipt with these details has been sent to the email on file.

12.2 Notification Preferences

- The Agent must support dynamic channel selection based on user input.
- **Default:** Both Email and SMS are enabled.
- **Opt-Out Logic:**
  - If user says *"Don't text me"*, the Agent sets `sms_opt_in = FALSE`.
  - If user says *"Email is fine"*, the Agent sets `email_opt_in = TRUE` and `sms_opt_in = FALSE` (if implied).

12.3 Proactive Reminders

- **Trigger:** During the closing phase of a Move/Install order.
- **Agent Script:** *"Would you like a reminder the day before to prep for the install?"*
- **Logic:**
  - **Opt-In:** Schedule a notification job for `[Move_Date - 1 Day]` at 10:00 AM.
  - **Opt-Out:** Record preference and close without scheduling.

13. Graceful Handling & Edge Cases

13.1 Identity & Context Errors

- **Invalid Account ID:**
  - **Trigger:** User enters an ID not in the "Cheat Sheet" (e.g., 55555).
  - **Action:** Agent prompts for retry. After 2 failures, it pivots to generic help.
- **Context Switching:**
  - **Trigger:** User authenticates with Account A, then says *"Wait, I gave you the wrong number. Let's use Account B."*
  - **Action:** The Agent must **flush** the current context/memory and restart the auth flow for Account B without crashing.
- **Redundant Cancellation:**
  - **Trigger:** User tries to cancel an account with `Status = CANCELED`.
  - **Action:** Agent informs the user the account is already closed and asks if they need anything else.

13.2 Address & Serviceability Errors

- **Out-of-Bounds:**
  - **Trigger:** Address not in the "Master List" (Section 2.2).
  - **Action:** Return specific error: *"I do not recognize that address. Please provide a valid Metro City address."*.
- **The "Same Address" Trap:**
  - **Trigger:** User enters their current address as the "New Address".
  - **Action:** Agent catches the duplicate and asks for the *destination* address.

13.3 Ambiguous Inputs

- **Ambiguous Consent:** If a user answers a Payment confirmation with *"I guess"* or *"Maybe"*, the Agent must seek explicit confirmation (*"Please answer Yes or No"*) before charging the card.

13.4 User Journey for Order Status ("The Pivot")

- Since we are not building a dedicated "Order Retrieval" database, the Agent handles status inquiries by pivoting back to the **Account ID**.

- **Scenario:** User asks, *"What is the status of my order #ORD-8821X?"*
- **Agent Logic:**
    - **Recognize Intent:** The Agent identifies the user is asking about a past request.
    - **Redirect (The Pivot):** The Agent explains that status is verified via the Account ID.
    - **Script:** *"I can check the current status of your service for you. For security, please provide your 5-digit Account ID."*
- **Outcome:** The user provides the Account ID, and the Agent confirms, *"Yes, Account 10001 is currently Active at 100 First St,"* effectively confirming the order was processed successfully.

14. Out-of-Scope & Unsupported Scenarios

- This section defines the "Hard Stops" where the Agent must explicitly decline service. For any request listed below, the Agent must use the **Standard Handoff Phrase**: "*I'm not able to complete this request in the virtual assistant. A human agent can help you with this. Have a nice day. Please continue with VA after your chat with the agent."*
- **Hardware Troubleshooting:** The Agent cannot diagnose slow speeds or Wi-Fi issues. It assumes service is working.
- **Payment Method Updates:** The Agent cannot update Credit Card numbers or expiration dates (Security restriction). It can only use the card on file.
- **Bundles & Promotions:**
    - **Scope Constraint:** The Agent does not support TV, Wireless, or Voice bundles. It cannot negotiate monthly rates or offer "special retention offers" beyond the standard catalog prices.
    - **Exception (The "Waiver Pivot"):** If a user explicitly asks for "promotions," "discounts," or "deals," the Agent **should not** use the hard stop. Instead, it must pivot to checking eligibility for the **Installation Fee Waiver (Tool T8)**.
        - *Script:* "While our monthly rates are fixed, I can check if you qualify to have the $99 installation fee waived. Let me check that for you."
- **Multi-Location Accounts:** The Agent does not support adding a second active line. The logic is strictly "One Account = One Address."